01 / SERVICE ECONOMICS

Define one service unit before doing arithmetic.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Unit

One bounded review of a passage and source question.

Duration assumption

45 teaching minutes per completed session.

Exclusion

No guarantee of correctness or full-document approval.

What must be clear before a request can count as one service unit?

The fictional service unit is one bounded document-review session lasting forty-five teaching minutes. It examines a passage and its source question, not the correctness of every claim in a document.

Specify what is included and what remains outside the unit. This workbook establishes no operating service, professional verification, appointment availability or commercial contract. A real service definition needs appropriate responsibility and scope.

Unit definition

The model uses one arbitrary planning window. Neither the duration nor the resource assumptions are validated observations of actual work.



Figure 1 / fictional illustration, not evidence / curated image source in provenance.json

02 / SERVICE ECONOMICS

Keep demand assumptions separate from bookings.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Lower case

6 hypothetical requests.

Reference case

8 hypothetical requests.

Higher case

10 hypothetical requests, not actual bookings.

Which definition would distinguish a request from a confirmed suitable booking?

The three model demand cases contain six, eight and ten requested sessions. They are authored planning inputs, not a waiting list, booked work or a measure of customer interest.

An actual demand assessment would need a defined request, a relevant window and an appropriate source. Counting enquiries as completed work would conceal both uncertainty and the delivery constraint.

Demand register

Demand above the model’s capacity produces an unmet count, not automatic permission to shorten sessions or make commitments.



Figure 1 / fictional illustration, not evidence / curated image source in provenance.json

03 / SERVICE ECONOMICS

Derive capacity from the stated time assumptions.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Available / reserve

450 minutes less 90 reserved = 360.

Unit / capacity

360 ÷ 45 = 8 whole sessions.

Feasibility boundary

No observed duration, staff availability or workload approval.

Which assumption must be checked before calling eight sessions feasible?

The model starts with four hundred and fifty available teaching minutes and reserves ninety. That leaves three hundred and sixty minutes for the defined service units.

At forty-five minutes each, floor of 360 divided by 45 gives eight sessions. The reserve is a modelling boundary, not spare time that the workbook authorises someone to consume. No actual staffing arrangement is supplied.

Capacity calculation

Changing a unit’s duration or the reserve changes the model. Preserve the original assumption and examine whether the service scope would still be appropriate.



Figure 2 / fictional illustration, not evidence / curated image source in provenance.json

04 / SERVICE ECONOMICS

Map the work around the session.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Before	During	After
Clarify scope and source availability.	Review the bounded passage-question relationship.	Record the remaining question under an appropriate agreement.

Which supporting task could make the simple capacity assumption misleading?

Before a session, clarify the passage and relevant source. During it, inspect the stated question. After it, retain an appropriately agreed note about what remains unresolved.

The time model does not separately measure preparation, coordination or follow-up. If those activities draw on the same available minutes, the capacity calculation must be revised rather than quietly treating them as free.

Service workflow

The ninety-minute reserve is not an itemised estimate of all supporting work. Real delivery needs a more complete account of tasks and responsibilities.



Figure 2 / fictional illustration, not evidence / curated image source in provenance.json

05 / SERVICE ECONOMICS

Use a unit model without pretending it is money.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Unit contribution

10 − 4 = 6 teaching credits.

Window net

6 × completed sessions − 20 fixed credits.

Teaching threshold

ceil(20 ÷ 6) = 4 sessions; not financial break-even.

What evidence would be needed before constructing an actual commercial unit model?

Each completed session receives ten benefit credits and uses four variable resource credits. The unit contribution is six credits; fixed resources are twenty credits per teaching window.

Credits are arbitrary non-monetary quantities used to illustrate structure. Net credits equal six times completed sessions minus twenty. They are not prices, sales, wages, cash flow or profit.

Unit model / teaching credits

The smallest whole number covering the model’s fixed credits is ceiling of 20 divided by 6, which is four completed sessions. This is a teaching threshold, not a real financial break-even.



Figure 3 / fictional illustration, not evidence / curated image source in provenance.json

06 / SERVICE ECONOMICS

Distinguish fixed assumptions from fixed obligations.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Model assumption

20 fixed teaching credits per window.

Time boundary

90 minutes reserved in the capacity model.

Missing costs

No actual spending, rates or contractual obligations supplied.

Which supposedly fixed resource might change when the delivery arrangement changes?

The model holds twenty fixed credits and a ninety-minute reserve unchanged across the three cases. That is a comparison choice, not evidence of an actual obligation or a complete operating cost base.

A real service may have resources that change in steps, shared capacity or commitments outside the planning window. Identify those conditions before translating a simple unit model into a decision.

Fixed-resource questions

There are no rent, salary, tax, technology or insurance figures in this workbook. Do not interpret the teaching total as an actual cost estimate.



Figure 3 / fictional illustration, not evidence / curated image source in provenance.json

07 / SERVICE ECONOMICS

Compare the three demand cases.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Lower case

6 delivered; 90 unused service minutes; net 16.

Reference case

8 delivered; 0 unused service minutes; net 28.

Higher case

8 delivered; 2 unmet requests; net remains 28.

Which total stops changing when the capacity ceiling is reached?

Demand six, eight and ten produces delivery six, eight and eight under the capacity ceiling. Unmet counts are zero, zero and two. Unused service minutes are ninety, zero and zero.

Net teaching credits are sixteen, twenty-eight and twenty-eight. The higher demand case cannot raise delivery or net credits without changing another assumption. No actual utilisation or financial performance is reported.

Constrained case comparison

Service minutes are delivered sessions times forty-five. Unused service minutes exclude the ninety-minute reserve, which remains separate in every case.



Figure 4 / fictional illustration, not evidence / curated image source in provenance.json

08 / SERVICE ECONOMICS

Treat quality as a condition, not an automatic output.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Scope check

The task remains bounded and understandable.

Support check

Relevant access and review conditions are considered.

Evidence gap

No actual service-quality or duration observations supplied.

What could be lost if the unit were compressed without changing its scope?

Completing eight units in the model says nothing about the suitability or quality of an actual review. A capacity equation cannot establish whether a participant had enough context or support.

Do not shorten a session merely to make the arithmetic fit more requests. Consider task complexity, access arrangements and the agreed review boundary before changing the unit definition or reserve.

Quality and capacity trade-off

The workbook provides no quality score, satisfaction result or validated duration. Any real trade-off requires relevant information and appropriate responsibility.



Figure 4 / fictional illustration, not evidence / curated image source in provenance.json

09 / SERVICE ECONOMICS

Build a validation checklist around the unknowns.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Scope evidence

An agreed definition of the service and exclusions.

Time evidence

Appropriate observations of the full task, not just contact time.

Commercial evidence

Actual terms and costs; none supplied here.

Which unknown would most change the interpretation of the model?

Before using a similar model, examine the request definition, actual task duration, supporting work, appropriate reserve and commercial conditions. Each needs a source rather than a confident guess.

Validate the meaning of the unit before seeking a precise number. A carefully calculated result can still be irrelevant if the service scope or planning window has changed.

Validation sequence

Start with scope and responsibility, then inspect time and demand information. Only use actual financial inputs when they are available and appropriately defined.



Figure 5 / fictional illustration, not evidence / curated image source in provenance.json

10 / SERVICE ECONOMICS

Turn the workbook into operating questions.

CAPACITY MODEL

(450 − 90) ÷ 45 = 8 whole sessions

UNIT MODEL

10 − 4 = 6 credits / ceil(20 ÷ 6) = 4; not financial break-even.

DEMAND / CAPACITY / UNIT

Scope question

What work is included and appropriately supported?

Resource question

What actually constrains delivery in the relevant setting?

Demand question

Which requests represent a suitable, defined need?

What should be learned before the service concept is developed further?

The useful questions are whether the unit is meaningful, whether the time boundary is defensible and what happens when suitable requests exceed capacity. The point model cannot answer those questions alone.

This document remains a fictional planning example. No actual service, pricing decision, staffing approval or profitability claim follows from its calculations. Record which source would be needed for a more consequential decision.

Next planning discussion

Carry forward one uncertainty about scope, one about resources and one about demand. Keep the teaching calculation available for inspection without upgrading it into a forecast.



Figure 5 / fictional illustration, not evidence / curated image source in provenance.json